

Ryan Huang

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TECHNICAL SKILLS

Programming Languages: C/C++, Python, Bash, Java, MATLAB, R
Embedded & Firmware Systems: Arduino, STM32, ESP32, Raspberry Pi, Serial Protocols (UART, I²C, SPI)
Networking & Communication: Network Protocols (IP-based, TCP/UDP, DNS, DHCP), Wireshark, OCPP
Development Tools & Environments: Linux, Git, Docker, VS Code, REST APIs, JSON, Jira, Altium

EDUCATION

The University of British Columbia Vancouver, BC
Bachelor of Applied Science (BASc) in Biomedical Engineering Expected Graduation: May 2028
Relevant Courses: Control Systems, Circuits & EM, Biomedical Instrumentation, MEMS Devices, Software Construction

WORK EXPERIENCE

Firmware Developer Co-op Jan. 2026 - Aug. 2026
Corinex Vancouver, BC

- Successfully deployed and installed the first broadband over power line (BPL) device in North America, bridging EV charger infrastructure with smart grid communication
- Designed an automated OCPP testing pipeline in Python using HTTP requests and command terminal which reduced manual validation time by 90%
- Captured TCP/IP network traffic packets and performed Wireshark analysis to debug BPL integration with wireless LTE connectivity and OCPP protocol
- Wrote and executed shell scripts on embedded systems (BusyBox) to compare PHY speed, device uptime and LTE signal strength (RSSI), reporting 30% signal fluctuations which contributed to future deployment plans

DESIGN TEAM EXPERIENCE

Electrical & Embedded Team Lead Sep. 2025 - Present
UBC Biomedical Engineering Student Team (BEST) Vancouver, BC

- Led the electrical team in designing the circuit framework for a multi-motor finger stretching device for a quadriplegic
- Responsible for programming an Arduino microcontroller to abstract user-controlled stretching into PWM, evaluating stretch angles to pulse speeds between 500-2500 μ s
- Developed a feedback loop system to monitor exercise conditions reducing mechanical failure rates by 90%
- Contributed to the preparation of an abstract that was peer-reviewed and accepted for publication in the *Gerontechnology* Journal (International Society for Gerontechnology) [Link to Abstract](#)

TECHNICAL PROJECTS

Embedded Game Boy APU Emulator - Project In Progress

- Built an embedded Game Boy APU emulator in C++, using timer interrupts and PWM signal generation to synthesize retro 8-bit music on STM32 hardware
- Modeled and programmed the Game Boy's 4-channel audio architecture by configuring frequency, duration, and duty cycle to synchronize real-time audio mixing
- Designed an embedded audio SDK on GitHub featuring reusable firmware APIs and sample song modules, enabling users to deploy custom multi-channel chiptune audio

Automated Sensor Monitor & Data Logger - Project Completed

- Built an end-to-end automation pipeline that redirected Arduino sensor data via a Windows COM-to-TCP bridge into Linux, parsed and timestamped values, flagged anomalies, generated plots, and automatically emailed reports
- Debugged and resolved cross-platform serial communication issues (baud-rate mismatches, COM permissions, TCP bridging), achieving 99% transmission reliability across 50+ test runs
- Automated embedded QA workflows by integrating Arduino firmware with Linux Bash scripting to streamline sensor data collection, anomaly detection, and visualization